Rishabh Sharma

Simons Center Independent Postdoctoral Fellow, New York University

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AI researcher and physicist using statistical physics to understand how AI models store information and build representations. Since joining NYU in July 2025, I have developed cyclic denoising, a memorization probe and extraction attack for diffusion models (**NeurIPS 2026, Main Track**), and Cross-View World Models (**ICLR 2026 Workshop on World Models**). I am first author and a corresponding author on both papers.

My doctoral research was published in ***Nature Physics*** and ***Physical Review Letters* (PRL)**, and my thesis received TIFR's **2026 Best Thesis in Physics** award. For the *Nature Physics* work, featured in *Nature Physics News & Views*, I also received the **2025 Ramakrishna Cowsik Medal**: best paper across the entire TIFR system, in any field, by a researcher under 35.

Interests: Mechanistic interpretability; AI alignment; AI for science; automated scientific discovery; world models.

Research Experience

Simons Center Independent Postdoctoral Fellow *Jul 2025 – present*
Simons Center for Computational Physical Chemistry, New York University

Cyclic Denoising Reveals Ultrastable Memories in Diffusion Models Paper | Landscape animations
NeurIPS 2026, Main Track

- **Invented cyclic denoising**, translating Ph.D. insights on memory and annealing in driven disordered systems into an extraction attack and dynamical probe. Controlled forward/reverse diffusion cycles reveal a **basin-hopping picture** of the generative landscape: memorized images occupy ultrastable basins, persist for **thousands of cycles**, and regenerate after near-total corruption.
- Efficiently **recovers memorized training images** even when training captions are unavailable, a setting where caption-dependent extraction attacks cannot operate. Finds candidates in **Stable Diffusion v1.4 and a pixel-space DDPM** through long-time dynamics with sampler-level control, requiring no prompts, prior training-data knowledge, auxiliary detectors, gradients, or weight inspection.

Cross-View World Models Paper | Project & demos
ICLR 2026 Workshop on World Models

- Led a collaboration with NYU neuroscientists and industry partner Statespace. Introduced an objective to predict future observations from another viewpoint, training viewpoint- and action-conditioned **diffusion transformers** on 4,186 one-minute Aimlabs episodes from four synchronized cameras.
- Identified a **view-invariant "cognitive map"** in middle transformer layers: linear pose probes transfer between egocentric and bird's-eye views without retraining. Yaw interventions rigidly rotate its heading ring by the commanded angle, leaving earlier, view-dependent heading unchanged.

Doctoral Researcher *2019 – 2025*
Tata Institute of Fundamental Research (TIFR), Hyderabad, India

- Discovered a correspondence between activity-induced and cyclic-shear annealing in amorphous solids, linking annealing to a ductile-to-brittle transition (***Nature Physics***, 2025; featured in *News & Views*).
- Developed fast, fault-tolerant mechanical-memory protocols (***Physical Review Letters***, 2025; co-first author). Conducted large-scale molecular-dynamics simulations using **C and MPI** on HPC clusters.

Technical Skills

Machine learning: Python, PyTorch, Hugging Face Diffusers; DDPM/DDIM, latent VAEs, U-Nets, DiTs.

Training & HPC: Multi-GPU H100/H200 training; Slurm, Apptainer/Singularity; C/MPI; 7+ years in HPC.

GPU programming: Built Numba CUDA kernels for molecular dynamics (Roskilde University, Denmark).

Publications

[1] **R. Sharma**, S. Martiniani. *Cyclic Denoising Reveals Ultrastable Memories in Diffusion Models.* **NeurIPS 2026, Main Track**. arXiv:2606.24000.
First and corresponding author: R. Sharma.

[2] **R. Sharma**, G. Hogervorst, W. E. Mackey, D. J. Heeger, S. Martiniani. *Cross-View World Models.* **ICLR 2026 Workshop on World Models**. arXiv:2602.07277.
First and co-corresponding author: R. Sharma.

[3] **R. Sharma**, S. Karmakar. *Activity-induced annealing leads to a ductile-to-brittle transition in amorphous solids.* **Nature Physics** (2025).
Featured in *Nature Physics News & Views*.

[4] M. Adhikari*, **R. Sharma***, S. Karmakar. *Encoding fast and fault-tolerant memories in bulk and nanoscale amorphous solids.* **Physical Review Letters** (2025).

[5] U. A. Dattani*, **R. Sharma***, S. Karmakar, P. Chaudhuri. *Cavitation instabilities in amorphous solids via secondary mechanical perturbations.* arXiv preprint (2023). arXiv:2303.04529.

*Co-first authors.

Education

Integrated Ph.D. in Physics *2019 – 2025*
Tata Institute of Fundamental Research (TIFR), Hyderabad, India
Advisor: Prof. Smarajit Karmakar.
Thesis: *Annealing and Memory Effects in Active and Passive Amorphous Solids.*

B.E. in Mechanical Engineering *2014 – 2018*
Panjab University, Chandigarh, India

Awards & Fellowships

Geeta Udgaonkar Award for Best Thesis in Physics, TIFR Alumni Association 2026

Ramakrishna Cowsik Medal, TIFR, for the *Nature Physics* paper 2025

Simons Center Independent Postdoctoral Fellowship, NYU 2025 – 2027
Fully funded fellowship with independence in research direction.

Sarojini Damodaran International Student Travel Fellowship, TIFR 2023
Supported research stays at Roskilde University and Heinrich Heine University Düsseldorf.

Talks & Presentations

NeurIPS 2026, poster, Sydney, Australia Dec 2026
New York University, Simons Center Postdoctoral Research Symposium, talk, USA Oct 2026
ICLR 2026 Workshop on World Models, poster, Rio de Janeiro, Brazil Apr 2026
APS Global Physics Summit, contributed talk, Denver, USA Mar 2026
New York University, Simons Center Postdoctoral Research Symposium, talk, USA Oct 2025
University of Göttingen, Institut für Theoretische Physik, seminar, Germany Jan 2024
University of Amsterdam, Computational Soft Matter Seminar, talk, Netherlands Jan 2024
Heinrich Heine University Düsseldorf, Soft Matter Group seminar, Germany Jan 2024
Roskilde University, "Glass and Time" Group seminar, Denmark Nov 2023

Professional Service

Journal reviewer: PRX Intelligence; Physical Review Materials (PRM).

Workshop reviewer: NeurIPS 2026 Workshop on Physical Understanding for Decision-Making: Bridging Foundation Models and Reliable Agents.